

Week 2 Reflection
TCSS-498 Directed Reading (Raghavi Sakpal)
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A quick review of Forward Kinematics (FK): A frame has a coordinate system defined by the axis of joint rotation (z), the direction toward the next connecting frame (x), and the axis orthogonal to both x and z (y). The n-th frame's coordinate system is related to the n-1 frame by projection onto it. This projection is the basis for the rotation matrix, which is then multiplied by the joint axis's standard rotation matrix. That rotation matrix is multiplied by the displacement vector to give each frame's position relative to the previous one in space. Finally, the transformation matrix is the sequential product of each frame's matrix in the chain, providing the end-effector's position in 3D space.

With Inverse Kinematics (IK) the process works in reverse, using the end-effector's position to determine the angles each joint must take to attain or maintain a given position. In FK there was a single unique solution (end-effector position), but in IK multiple solutions can exist for a single point in space. The two main methods for approaching the IK problem are the Geometric and the Algebraic approaches.

The Geometric Approach uses a process called kinematic decoupling to break the total problem down into two smaller ones. The dividing point is the wrist joint — the joint that rotates the end-effector. Inverse Position Kinematics (IPK) determines the joint configuration that places the wrist center at its ideal position in space. Inverse Orientation Kinematics (IOK) then resolves the remaining joint angles from the wrist outward, treating each joint angle as the interior angle of a triangle formed by the connecting links. This process requires the exact link lengths to calculate the required joint angles. The Geometric Approach is relatively fast compared to the Algebraic method, but it doesn't translate well between different robot structures. For my implementation I will start with the Geometric Approach and, if time allows, implement the Algebraic Approach as a check.

The Algebraic Approach is essentially an inversion of the matrix multiplication process and demands significantly more attention to the math during implementation compared to the repeated trig functions of the Geometric Approach. I'm still working through the details of the Algebraic Approach, and at first glance it would require a more adaptable algorithm to handle multiple solutions. Both methods, however, will require a separate selection algorithm to choose between valid solutions. It is also worth noting that while the Geometric Approach is straightforward in its process, the Algebraic framework can also accommodate other solving methods such as the Jacobian Inverse and Newton-Raphson. All of these would be great to validate against each other if time permits.

As a status update, I'm using the SO-ARM101 leader/follower arm design (link [here](#)) for this project. This is a well-developed hobbyist kit; however, I'm going to use an ESP32 instead of an Arduino board to avoid any project-specific prebuilt libraries. The servos used in the arms communicate over Transistor-Transistor Logic (TTL), a simple protocol that allows a single signal line to be shared across all servos. While the protocol itself is straightforward and implementing a pair of parsing functions for servo ID and motor signaling is manageable, my main concern is the translation between the servo adapter board (half-duplex) and the ESP32 (simplex). A secondary protocol I need to address is ESP-NOW, which will allow the two ESP32s (leader/follower) to communicate wirelessly. Fortunately, ESP-NOW is heavily documented, requires no encryption or security configuration, and operates essentially as plain text.

For handling robotic motion, I have three things I'm looking to test: teleoperation, Forward Kinematics, and Inverse Kinematics. For teleoperation I will simply mirror the leader's joint angles to the follower's servos. This is a straightforward translation that primarily serves to verify all communication protocols are functioning and data is being logged correctly. Once that is established, I will implement FK on the leader and IK on the follower and run a waypoint-style exercise. All these tests can be run live and will display corresponding telemetry data for each arm and joint in a terminal window.